

👉 CS301 = Data Structures

iska matlab:

data ko smart tareeke se store + fast access + kam time me kaam

♦ CS301 FINAL TERM = KIS CHEEZ SE AATA HAI?

Mostly ye topics aate hain (aur paper yahin se banta hai):

- 1 Trees (Binary Tree, BST)
- 2 Sorting (Bubble, Selection, Insertion, Merge)
- 3 Searching (Binary Search)
- 4 Heap
- 5 Hashing
- 6 Graphs (thora sa)

👉 Main ab ek road map follow karunga:

- Pehle **basic concept**
 - Phir **exam wali baat**
 - Phir **short trick ya formula**
-

✅ TOPIC 1: TREE KYA HOTA HAI? (Basic se)

♦ Tree ka matlab?

Socho **family tree** 🌳

- upar **father**
- neeche **children**

Computer me bhi same cheez hoti hai.

♦ Binary Tree kya hota hai?

Binary ka matlab = **sirf 2**

➡ Har node ke:

- ya to **0**
 - ya **1**
 - ya **2 children** hotay hain
2 se zyada nahi
-

◆ Node kya hota hai?

Node = ek **box** jisme:

- data hota hai
 - pointers hotay hain (left, right)
-

◆ Root kya hota hai?

Tree ka **sab se upar wala node** = **Root**

◆ Leaf node kya hota hai?

Jis node ke **koi bachay nahi hotay**

➡ usay kehte hain **Leaf / External Node**

◆ Internal node kya hota hai?

Jis node ke **bachay hotay hain**

➡ usay kehte hain **Internal Node**

★ EXAM GOLD FORMULA (YAAD KAR LO)

Binary Tree me:

External Nodes = Internal Nodes + 1

Example:

- Internal = 50
- External = 51

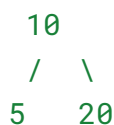
→ Guaranteed MCQ / short question

✓ TOPIC 2: BINARY SEARCH TREE (BST)

BST = special binary tree

♦ Rule (bohot important):

- Left side = **chota**
- Right side = **bara**



$5 < 10$

$20 > 10$

→ Is rule ki wajah se **search fast** hoti hai

♦ BST me searching fast kyun?

Kyunke:

- har step pe **half data reject**
-



TOPIC 3: SORTING (Paper ka favorite)

Sorting = numbers ko order me lana

◆ Bubble Sort (sab se easy)

◆ Logic:

- pass pass walay compare
 - bara element end pe chala jata hai
 - isliye naam bubble (bubble upar jata hai 🫧)
-

Example:

Array:

20, 8, 5, 10

First Pass:

① 20 & 8 → swap
→ 8, 20, 5, 10

② 20 & 5 → swap
→ 8, 5, 20, 10

③ 20 & 10 → swap
→ 8, 5, 10, 20

➡ 20 end pe chala gaya

◆ Bubble Sort exam line:

Largest element har pass ke baad end pe chala jata hai.



TOPIC 4: BINARY SEARCH (VERY IMPORTANT)

Binary Search sirf tab kaam karta hai jab:

👉 array sorted ho

♦ Logic:

- middle element check karo
 - agar value choti ho → left side
 - agar bari ho → right side
-

♦ Formula:

`mid = (low + high) / 2`

♦ Agar:

`val < arr[mid]`

➡ matlab value **left side** me hogi

`high = mid - 1`

♦ Exam wali line:

Since array sorted hota hai, isliye lower half choose hoti hai.



TOPIC 5: HEAP (Array + Tree)

Heap = Complete Binary Tree

♦ **Complete Binary Tree matlab:**

- sab levels full
 - last level **left se fill**
-

♦ **Heap array me store hota hai**

Index **1 se start** hota hai (important)

♦ **Formula yaad rakhna (100% exam):**

Agar node index = i

Parent = $i / 2$

Left Child = $2i$

Right Child = $2i + 1$

Example:

$i = 7$

- Parent = 3
 - Left = 14
 - Right = 15
-



AB REAL TALK (EXAM STRATEGY)

👉 Tumhara target:

- **MCQs easy pick**
- **Section B = ye topics**

- **Definitions + formula + 2 lines logic**

👉 Writing style:

- Headings
- Formula
- Short explanation